

CIL MT 2026 — FULL-LENGTH MOCK CBT 1

Coal India Limited | Management Trainee | System Discipline (Post Code 14) | Advt. No. 03/2026

Total questions: 200 | **Total marks:** 200 | **Total time:** 3 hours (180 minutes), single sitting

Paper-I — General Knowledge/Awareness, Reasoning, Numerical Ability, General English	Q1 – Q100 (100 marks)
Paper-II — Professional Knowledge, System (CS/IT)	Q101 – Q200 (100 marks)

Instructions — follow these exactly to make the practice count

1. Set a single timer for 180 minutes and do not pause it. Both papers are attempted in one sitting, exactly as in the CBT.
 2. Each question carries 1 mark. **There is no negative marking** — leave nothing blank. Mark your best guess on anything you cannot solve.
 3. Both papers must be qualified **separately**: 40 marks for UR/EWS, 35 for OBC-NCL, 30 for SC/ST/PwD, out of 100 in each paper. Score each paper on its own before looking at the total.
 4. Suggested split: 75 minutes on Paper-I, 95 minutes on Paper-II, 10 minutes to sweep unattempted questions. Adjust after your first attempt.
 5. The answer key is at the end of this document. Do not open it until the timer stops.
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PAPER-I — GENERAL APTITUDE (Q1 - Q100)

SECTION A — GENERAL KNOWLEDGE / AWARENESS

- Q1.** The largest brackish water lagoon in India is
(A) Sambhar Lake (B) Vembanad Lake (C) Chilika Lake (D) Wular Lake
- Q2.** The words “Socialist” and “Secular” were added to the Preamble by which amendment?
(A) 42nd (B) 61st (C) 52nd (D) 44th
- Q3.** The Tropic of Cancer does NOT pass through which of the following states?
(A) Odisha (B) Tripura (C) Madhya Pradesh (D) Rajasthan
- Q4.** The speed of light in vacuum is approximately
(A) 3×10^{10} m/s (B) 3×10^5 m/s (C) 3×10^6 m/s (D) 3×10^8 m/s
- Q5.** The headquarters of the Reserve Bank of India is at
(A) Kolkata (B) Mumbai (C) New Delhi (D) Chennai
- Q6.** The largest producer of coal in the world is
(A) India (B) United States (C) Australia (D) China
- Q7.** The instrument used to record earthquakes is the
(A) seismograph (B) altimeter (C) barometer (D) hygrometer
- Q8.** The first Indian to be awarded a Nobel Prize was
(A) Mother Teresa (B) Rabindranath Tagore (C) C. V. Raman (D) Amartya Sen
- Q9.** South Eastern Coalfields Limited (SECL) operates mainly in
(A) Odisha and Jharkhand (B) West Bengal and Bihar (C) Maharashtra and Telangana (D) Chhattisgarh and Madhya Pradesh
- Q10.** The Bhakra Nangal project is built on which river?
(A) Beas (B) Sutlej (C) Ravi (D) Chenab
- Q11.** The number of Fundamental Rights presently guaranteed by the Indian Constitution is
(A) six (B) seven (C) five (D) eleven
- Q12.** Coking coal mines in India were nationalised in the year
(A) 1973 (B) 1971 (C) 1972 (D) 1975
- Q13.** The gas chiefly responsible for explosions in coal mines (firedamp) is
(A) methane (B) nitrogen (C) carbon dioxide (D) sulphur dioxide
- Q14.** The Indian Standard Meridian is
(A) $84^\circ 30'E$ (B) $82^\circ 30'E$ (C) $80^\circ 30'E$ (D) $85^\circ 30'E$
- Q15.** Safety in Indian mines is regulated by
(A) SEBI (B) DGMS (C) CPCB (D) BIS
- Q16.** The hardest naturally occurring substance is
(A) corundum (B) diamond (C) graphite (D) quartz
- Q17.** Raniganj, where commercial coal mining in India began in 1774, is located in
(A) Bihar (B) Odisha (C) Jharkhand (D) West Bengal
- Q18.** Northern Coalfields Limited (NCL) has its headquarters at
(A) Bhopal (B) Singrauli (C) Lucknow (D) Kanpur
- Q19.** The maximum permissible strength of the Rajya Sabha is
(A) 245 (B) 250 (C) 552 (D) 238
- Q20.** The Indian National Congress was founded in the year
(A) 1885 (B) 1905 (C) 1857 (D) 1919
- Q21.** A Money Bill is defined in which Article of the Constitution?
(A) Article 110 (B) Article 112 (C) Article 109 (D) Article 115
- Q22.** Coal India Limited is classified as
(A) a private limited company (B) a Miniratna CPSE (C) a Navratna CPSE (D) a Maharatna CPSE
- Q23.** Eastern Coalfields Limited (ECL) has its headquarters at
(A) Durgapur (B) Kolkata (C) Dhanbad (D) Sanctoria, West Bengal
- Q24.** The Right to Equality is guaranteed by which Article of the Indian Constitution?
(A) Article 32 (B) Article 14 (C) Article 21 (D) Article 19
- Q25.** The Election Commission of India derives its authority from which Article of the Constitution?
(A) Article 315 (B) Article 280 (C) Article 148 (D) Article 324

SECTION B — REASONING

- Q26.** If $A = 1$, $B = 2$, $C = 3$ and so on, the sum of the letters of the word DOG is
(A) 24 (B) 28 (C) 30 (D) 26
- Q27.** Find the next term: 6, 12, 24, 48, ?
(A) 72 (B) 96 (C) 144 (D) 108

- Q28.** Cobbler : Leather :: Carpenter : ?
(A) Furniture (B) Wood (C) Workshop (D) Hammer
- Q29.** In a certain code RAIN is written as SBJO. In the same code SNOW is written as
(A) TOQX (B) TOPX (C) TPOX (D) SOPX
- Q30.** Choose the odd one out.
(A) 7 (B) 9 (C) 3 (D) 5
- Q31.** Doctor : Patient :: Teacher : ?
(A) School (B) Lesson (C) Student (D) Book
- Q32.** Book : Author :: Painting : ?
(A) Canvas (B) Colour (C) Painter (D) Museum
- Q33.** The number of days in February 2024 is
(A) 28 (B) 29 (C) 30 (D) 27
- Q34.** If today is Monday, what day of the week will it be after 61 days?
(A) Saturday (B) Thursday (C) Friday (D) Sunday
- Q35.** Find the next term: 3, 6, 11, 18, 27, ?
(A) 42 (B) 36 (C) 40 (D) 38
- Q36.** Find the next term: 1, 1, 2, 3, 5, 8, ?
(A) 16 (B) 13 (C) 11 (D) 15
- Q37.** Find the next term: 11, 13, 17, 19, 23, ?
(A) 29 (B) 25 (C) 31 (D) 27
- Q38.** If MONDAY is coded as NPOEBZ, then FRIDAY is coded as
(A) GSJEBZ (B) ESHCZX (C) GSJFBZ (D) GSKEBZ
- Q39.** Find the next term: 2, 3, 5, 7, 11, ?
(A) 12 (B) 15 (C) 14 (D) 13
- Q40.** Complete the series: AZ, BY, CX, ?
(A) DV (B) DW (C) DX (D) EW
- Q41.** Choose the odd one out.
(A) Rectangle (B) Triangle (C) Circle (D) Square
- Q42.** A man walks 4 km north, turns right and walks 4 km, then turns right and walks 4 km. In which direction is he from his starting point?
(A) West (B) South (C) East (D) North
- Q43.** Find the next term: 5, 8, 14, 26, ?
(A) 38 (B) 44 (C) 52 (D) 50
- Q44.** A man walks 5 km north, turns right and walks 3 km, then turns right and walks 5 km. How far is he from his starting point?
(A) 5 km (B) 8 km (C) 13 km (D) 3 km
- Q45.** The angle between the hour hand and the minute hand of a clock at 3:30 is
(A) 90° (B) 75° (C) 80° (D) 60°
- Q46.** Find the next term: 2, 5, 10, 17, 26, ?
(A) 35 (B) 37 (C) 38 (D) 41
- Q47.** In a row of 40 children, A is 12th from the left end. What is A's position from the right end?
(A) 30th (B) 29th (C) 27th (D) 28th
- Q48.** A man walks 10 m north, then 10 m east, then 10 m south. He is now
(A) 10 m east of the starting point (B) 20 m from the starting point (C) 10 m north of the starting point (D) at the starting point
- Q49.** Choose the odd one out.
(A) Chennai (B) Kolkata (C) Bhopal (D) Mumbai
- Q50.** The angle between the hour hand and the minute hand of a clock at 6:00 is
(A) 90° (B) 120° (C) 150° (D) 180°

SECTION C — NUMERICAL ABILITY

- Q51.** The present ages of two persons are in the ratio 4 : 5. After 5 years the ratio becomes 5 : 6. The present age of the younger person is
(A) 24 years (B) 25 years (C) 20 years (D) 16 years
- Q52.** A can complete a work in 15 days and B in 10 days. Working together they will complete it in
(A) 7 days (B) 5 days (C) 8 days (D) 6 days
- Q53.** The value of 0.2×0.2 is
(A) 4.0 (B) 0.04 (C) 0.004 (D) 0.4
- Q54.** If $(x/5) + 3 = 8$, then x equals
(A) 25 (B) 40 (C) 15 (D) 55
- Q55.** A is twice as efficient as B. Working together they finish a job in 12 days. A alone would take
(A) 18 days (B) 36 days (C) 16 days (D) 24 days

- Q56.** The average of 10 numbers is 15. Their sum is
(A) 150 (B) 155 (C) 100 (D) 175
- Q57.** A boat has a speed of 10 km/h in still water and the current is 2 km/h. The time taken to travel 24 km downstream is
(A) 3 hours (B) 2 hours (C) 4 hours (D) 2.4 hours
- Q58.** A can complete a work in 20 days. If he works for 5 days and then leaves, the fraction of work completed is
(A) $\frac{3}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$
- Q59.** The price of a commodity is increased by 20% and then decreased by 20%. The net change is
(A) a 4% increase (B) no change (C) a 4% decrease (D) a 2% decrease
- Q60.** The simple interest on ₹8,000 at 5% per annum for 2 years is
(A) ₹900 (B) ₹700 (C) ₹1,000 (D) ₹800
- Q61.** If 10 men complete a work in 12 days, 15 men working at the same rate will complete it in
(A) 6 days (B) 18 days (C) 9 days (D) 8 days
- Q62.** The simple interest on ₹4,000 at 6% per annum for 6 months is
(A) ₹120 (B) ₹240 (C) ₹60 (D) ₹100
- Q63.** The compound interest on ₹5,000 at 20% per annum for 1 year, compounded half-yearly, is
(A) ₹1,050 (B) ₹1,000 (C) ₹1,100 (D) ₹1,200
- Q64.** If 25% of x is 50, then x equals
(A) 125 (B) 250 (C) 200 (D) 400
- Q65.** The compound interest on ₹5,000 at 10% per annum for 2 years is
(A) ₹1,050 (B) ₹1,150 (C) ₹1,100 (D) ₹1,000
- Q66.** An article bought for ₹400 is sold for ₹460. The profit percentage is
(A) 15% (B) 12% (C) 13% (D) 20%
- Q67.** A father is three times as old as his son and the sum of their ages is 48 years. The son's age is
(A) 14 years (B) 16 years (C) 10 years (D) 12 years
- Q68.** The HCF of 24 and 36 is
(A) 12 (B) 18 (C) 6 (D) 72
- Q69.** An amount of ₹900 is divided among three persons in the ratio 2 : 3 : 4. The largest share is
(A) ₹360 (B) ₹450 (C) ₹300 (D) ₹400
- Q70.** The volume of a cube of side 5 cm is
(A) 625 cm^3 (B) 150 cm^3 (C) 25 cm^3 (D) 125 cm^3
- Q71.** The area of a circle of radius 7 cm (taking $\pi = \frac{22}{7}$) is
(A) 154 cm^2 (B) 176 cm^2 (C) 308 cm^2 (D) 44 cm^2
- Q72.** If the side of a square is increased by 10%, its area increases by
(A) 10% (B) 25% (C) 20% (D) 21%
- Q73.** An article costing ₹600 is sold for ₹540. The loss percentage is
(A) 15% (B) 6% (C) 10% (D) 12%
- Q74.** Three-fourths of 240 is
(A) 180 (B) 200 (C) 160 (D) 120
- Q75.** An article costing ₹250 is sold at a loss of 12%. The selling price is
(A) ₹238 (B) ₹230 (C) ₹210 (D) ₹220

SECTION D — GENERAL ENGLISH

Directions (Q76-Q77): Read the following passage and answer the questions that follow.

A public sector enterprise is judged by two standards that do not always agree. As a commercial body it must return a profit to its shareholders, the largest of whom is the government itself. As an instrument of policy it is expected to keep fuel affordable, employ locally and invest in regions that private capital avoids.

- Q76.** The central idea of the passage is that a public sector enterprise
(A) must balance commercial and social objectives (B) has no shareholders (C) is always unprofitable (D) should be privatised
- Q77.** As used in the passage, "instrument of policy" suggests that the enterprise
(A) serves government objectives (B) sets its own policy freely (C) avoids all regulation (D) manufactures instruments

Directions (Q78-Q79): Read the following passage and answer the questions that follow.

Automation is often presented as a straightforward substitute for labour, but in practice it redistributes work rather than removing it. Machines require maintenance, calibration and supervision, and each of these creates a role that did not previously exist. What changes is not the quantity of work so much as the qualification required to perform it.

- Q78.** The author's main contention is that automation
(A) reduces machine maintenance (B) lowers wages universally (C) eliminates work entirely (D) shifts the type of work required
- Q79.** The word "redistributes" is used to suggest that work is
(A) moved into different roles (B) destroyed (C) made voluntary (D) doubled in volume

Directions (Q80-Q81): Read the following passage and answer the questions that follow.

Coal continues to supply the greater part of India's electricity even as solar capacity expands rapidly. Planners defend it on the ground that coal-fired stations deliver the steady baseload that intermittent renewables cannot yet guarantee. Critics reply that postponing investment in storage merely postpones the transition itself.

Q80. The passage is chiefly concerned with

- (A) the technical design of solar panels (B) the history of Indian coal mining (C) the cost of electricity to households (D) why coal retains a central role in India's power mix

Q81. In the passage, the word "intermittent" most nearly means

- (A) continuous and steady (B) inexpensive (C) occurring at irregular intervals (D) newly invented

Directions (Q82-Q83): Read the following passage and answer the questions that follow.

Interviews reward confidence and articulation, qualities that correlate imperfectly with competence. A candidate who has rehearsed answers may outperform one who has done the work but speaks plainly. This is not an argument against interviews, only against treating them as the final word.

Q82. The passage suggests that interviews

- (A) are cheaper than written tests (B) should be abolished altogether (C) always identify the best candidate (D) can favour presentation over ability

Q83. The phrase "correlate imperfectly with" means

- (A) are identical to (B) are only partly related to (C) are opposed to (D) are entirely unrelated to

Q84. One who cannot be corrected is called

- (A) incorrigible (B) inevitable (C) illegible (D) invincible

Q85. Choose the correct indirect form of: He said, "I am busy."

- (A) He said that I am busy (B) He says that he is busy (C) He said that he is busy (D) He said that he was busy

Q86. Choose the word most similar in meaning to ELOQUENT.

- (A) Fluent (B) Rude (C) Silent (D) Hesitant

Q87. Choose the word most similar in meaning to ABUNDANT.

- (A) Scarce (B) Plentiful (C) Delicate (D) Hidden

Q88. Choose the word most opposite in meaning to ANCIENT.

- (A) Sacred (B) Modern (C) Old (D) Historic

Q89. Identify the part of the sentence containing an error: "He is one of / the best player / in the team. / No error"

- (A) in the team (B) He is one of (C) the best player (D) No error

Q90. Choose the word most similar in meaning to PRUDENT.

- (A) Cautious (B) Reckless (C) Stubborn (D) Wealthy

Q91. Choose the word most opposite in meaning to HOSTILE.

- (A) Angry (B) Cautious (C) Aggressive (D) Friendly

Q92. Choose the correct passive form of: "The manager approved the proposal."

- (A) The proposal was approving by the manager (B) The proposal was approved by the manager (C) The proposal is approved by the manager (D) The proposal has approved by the manager

Q93. A speech delivered without previous preparation is called

- (A) verbose (B) eloquent (C) extempore (D) rehearsed

Q94. Choose the word most similar in meaning to TERMINATE.

- (A) End (B) Begin (C) Delay (D) Continue

Q95. Identify the part containing an error: "She is senior / than me / in this office. / No error"

- (A) in this office (B) than me (C) She is senior (D) No error

Q96. Choose the word most opposite in meaning to VICTORY.

- (A) Triumph (B) Glory (C) Defeat (D) Struggle

Q97. Identify the part containing an error: "One of my friend / is a doctor / in this city. / No error"

- (A) in this city (B) One of my friend (C) is a doctor (D) No error

Q98. The idiom "to be in hot water" means

- (A) to be very busy (B) to be in trouble (C) to be angry (D) to be in good health

Q99. Choose the word most opposite in meaning to ABUNDANCE.

- (A) Wealth (B) Scarcity (C) Variety (D) Plenty

Q100. A place where coins are manufactured is called a

- (A) granary (B) mint (C) treasury (D) foundry

SECTION 1 — ENGINEERING MATHEMATICS

- Q101.** The number of distinct arrangements of the letters of the word "LEVEL" is
(A) 60 (B) 20 (C) 30 (D) 120
- Q102.** For events A and B with $P(B) > 0$, the conditional probability $P(A | B)$ equals
(A) (B) $P(A \cap B)/P(A)$ (B) $P(A) \cdot P(B)$ (C) $P(A \cap B)/P(B)$ (D) $P(A \cup B)/P$
- Q103.** Near a simple root, the Newton-Raphson method converges
(A) linearly (B) cubically (C) quadratically (D) not at all
- Q104.** The number of distinct labelled spanning trees of the complete graph K_4 is
(A) 64 (B) 16 (C) 4 (D) 8
- Q105.** The eigenvalues of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ are
(A) 1 and 3 (B) 1 and 2 (C) 2 and 3 (D) 0 and 3
- Q106.** A tree has 10 edges. The number of vertices in the tree is
(A) 11 (B) 12 (C) 10 (D) 9
- Q107.** The determinant of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is
(A) 0 (B) 24 (C) -6 (D) 1
- Q108.** For a symmetric matrix A
(A) A is always orthogonal (B) $A^T = -A$ (C) $A^T = A$ (D) A is always singular
- Q109.** The number of reflexive relations that can be defined on a set of 4 elements is
(A) 2^{16} (B) 2^4 (C) 2^{10} (D) 2^{12}
- Q110.** The sum of all eigenvalues of a square matrix is equal to its
(A) trace (B) determinant (C) norm (D) rank
- Q111.** The standard deviation of the data 2, 4, 4, 4, 5, 5, 7, 9 is
(A) 1 (B) 4 (C) 2.5 (D) 2
- Q112.** Simpson's $1/3$ rule for numerical integration requires the number of subintervals to be
(A) odd (B) prime (C) even (D) any positive integer

SECTION 2 — DIGITAL LOGIC

- Q113.** The Boolean expression $A'B + AB$ simplifies to
(A) AB (B) B (C) A (D) $A + B$
- Q114.** The number of 2-input NAND gates required to realize a NOT gate is
(A) 3 (B) 4 (C) 2 (D) 1
- Q115.** In the IEEE 754 single-precision format, the number of bits allotted to the exponent is
(A) 8 (B) 7 (C) 11 (D) 23
- Q116.** The number of distinct states in a MOD-16 counter is
(A) 15 (B) 4 (C) 8 (D) 16
- Q117.** The dual of the Boolean expression $A + (B \cdot C)$ is
(A) $A \cdot (B + C)$ (B) $A \cdot (B \cdot C)$ (C) $A + (B + C)$ (D) $A' + (B' \cdot C')$
- Q118.** The range of integers representable in 8-bit 2's complement form is
(A) 0 to 255 (B) -127 to +127 (C) -128 to +127 (D) -128 to +128
- Q119.** The hexadecimal equivalent of the binary number 11011010 is
(A) AD (B) DA (C) D8 (D) BD
- Q120.** A full adder can be realized using
(A) two half adders and one OR gate (B) one half adder and one AND gate (C) two half adders only (D) three half adders

SECTION 3 — COMPUTER ORGANIZATION & ARCHITECTURE

- Q121.** Virtual memory in modern operating systems is normally implemented using
(A) overlays only (B) demand paging (C) contiguous allocation (D) swapping of entire processes only
- Q122.** A memory chip organized as $1K \times 8$ requires
(A) 1024 address lines, 8 data lines (B) 10 address lines, 8 data lines (C) 8 address lines, 10 data lines (D) 10 address lines, 1 data line
- Q123.** The interrupt vector table contains
(A) interrupt data (B) only interrupt priorities (C) starting addresses of interrupt service routines (D) opcodes of interrupt instructions
- Q124.** Translation of a logical address into a physical address at run time is performed by the
(A) ALU (B) compiler (C) DMA controller (D) memory management unit
- Q125.** The register that always holds the address of the next instruction to be fetched is the
(A) Accumulator (B) Program Counter (C) Instruction Register (D) MAR

- Q126.** In a direct-mapped cache with 1024 blocks and a block size of 8 bytes, the number of bits in the block-offset field of the address is
(A) 10 (B) 8 (C) 3 (D) 2
- Q127.** Instruction pipelining primarily improves
(A) the instruction throughput (B) the latency of an individual instruction (C) the size of main memory (D) the clock period only
- Q128.** The stack pointer register always contains
(A) the first instruction of the program (B) the address of the top of the stack (C) the interrupt vector (D) the base of the data segment
- Q129.** The number of address lines required for a 4 KB byte-addressable memory is
(A) 16 (B) 12 (C) 10 (D) 11
- Q130.** The instruction `MOV R1, #5` uses which addressing mode?
(A) Indexed (B) Indirect (C) Direct (D) Immediate
- Q131.** The correct order of operations in a basic instruction cycle is
(A) fetch, decode, execute (B) decode, fetch, execute (C) execute, fetch, decode (D) fetch, execute, decode
- Q132.** The fastest storage element in the memory hierarchy is
(A) CPU registers (B) main memory (C) L1 cache (D) solid-state disk

SECTION 4 — PROGRAMMING & DATA STRUCTURES

- Q133.** Which pair of traversals does NOT uniquely determine a binary tree?
(A) Preorder and postorder (B) Inorder and level order (C) Inorder and preorder (D) Inorder and postorder
- Q134.** In the array representation of a binary heap with the root at index 1, the children of the node at index i are at indices
(A) $2i - 1$ and $2i$ (B) $2i$ and $2i + 1$ (C) $i/2$ and $i/2 + 1$ (D) $i - 1$ and $i + 1$
- Q135.** The most space-efficient way to store a sparse matrix is
(A) a circular queue (B) a two-dimensional array (C) a triplet or linked representation of non-zero elements (D) a stack
- Q136.** In C, when an array is passed as an argument to a function, what is actually passed is
(A) a pointer to its first element (B) a copy of the entire array (C) a reference to the whole array object (D) the array size only
- Q137.** A circular queue implemented in an array of size n , keeping one slot always empty to distinguish full from empty, can hold at most
(A) $n - 1$ elements (B) $n/2$ elements (C) $n + 1$ elements (D) n elements
- Q138.** Inserting the sorted sequence 1, 2, 3, ..., n into an empty binary search tree produces a tree of height
(A) $\sqrt{n}$ (B) $\log_2 n$ (C) $n - 1$ (D) $n/2$
- Q139.** The minimum number of stacks required to implement a queue is
(A) 2 (B) 4 (C) 1 (D) 3
- Q140.** Depth-first search of a graph is naturally implemented using
(A) a stack (B) a priority queue (C) a hash table (D) a queue
- Q141.** In C, for the declaration `int a[5];` the expression `a[2]` is equivalent to
(A) `*(a + 2)` (B) `*a + 2` (C) `a + 2` (D) `&a + 2`
- Q142.** In C, the value of `sizeof(char)` is guaranteed to be
(A) machine dependent (B) 2 (C) 1 (D) 4
- Q143.** Given a pointer to a node in a doubly linked list, the time to delete that node is
(A) $O(n^2)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(1)$
- Q144.** To achieve $O(1)$ enqueue and $O(1)$ dequeue in a linked-list-based queue, the implementation must maintain
(A) only a head pointer (B) only a tail pointer (C) a doubly linked list, necessarily (D) both head and tail pointers

SECTION 5 — ALGORITHMS

- Q145.** Which of the following employ the divide-and-conquer strategy?
(A) Binary search only (B) Merge sort only (C) Quicksort only (D) All of merge sort, binary search and quicksort
- Q146.** The time complexity of the Bellman-Ford algorithm on a graph with V vertices and E edges is
(A) $O(VE)$ (B) $O(V^2)$ (C) $O(E \log V)$ (D) $O(V + E)$
- Q147.** Strassen's algorithm multiplies two $n \times n$ matrices in approximately
(A) $O(n^{2.81})$ (B) $O(n^3)$ (C) $O(n^2)$ (D) $O(n \log n)$
- Q148.** The minimum number of comparisons required to find both the minimum and the maximum of n elements is
(A) $2n - 2$ (B) $n - 1$ (C) $n \log n$ (D) $\lceil 3n/2 \rceil - 2$
- Q149.** Which algorithm selects the k th smallest element of an unsorted array in $O(n)$ worst-case time?
(A) Merge sort (B) Heapsort (C) Quickselect with a random pivot (D) Median-of-medians selection
- Q150.** The average-case time complexity of quicksort is
(A) $O(n \log n)$ (B) $O(n^2)$ (C) $O(n)$ (D) $O(\log n)$
- Q151.** The auxiliary space complexity of the standard merge sort on arrays is
(A) $O(\log n)$ (B) $O(n \log n)$ (C) $O(n)$ (D) $O(1)$
- Q152.** Matrix chain multiplication is solved optimally using
(A) divide and conquer (B) greedy method (C) dynamic programming (D) branch and bound

- Q153.** Which sorting algorithm runs in $O(n)$ time when the input consists of integers drawn from a small fixed range?
(A) Counting sort (B) Quicksort (C) Merge sort (D) Heapsort
- Q154.** The greedy approach does NOT yield an optimal solution for
(A) the fractional knapsack problem (B) the 0/1 knapsack problem (C) minimum spanning tree (D) Huffman coding
- Q155.** A topological ordering of vertices exists for
(A) directed graphs with cycles (B) any graph (C) any undirected graph (D) directed acyclic graphs only
- Q156.** The dynamic programming solution for the longest common subsequence of strings of lengths m and n takes
(A) $O(m + n)$ time (B) $O(n \log n)$ time (C) $O(mn)$ time (D) $O(2^n)$ time

SECTION 6 — THEORY OF COMPUTATION

- Q157.** The regular expression denoting all strings of even length over the alphabet $\{a\}$ is
(A) $a(aa)^*$ (B) $(a + aa)^*$ (C) $(aa)^*$ (D) a^*
- Q158.** The pumping lemma for regular languages is used to prove that a given language is
(A) decidable (B) not regular (C) context-free (D) regular
- Q159.** The halting problem for Turing machines is
(A) undecidable but recursively enumerable (B) decidable (C) not recursively enumerable (D) context-free
- Q160.** Context-free languages are closed under
(A) complementation (B) union (C) intersection (D) all of the above
- Q161.** The problem of deciding whether the language generated by a given context-free grammar is empty is
(A) not recursively enumerable (B) decidable (C) NP-complete but undecidable (D) undecidable
- Q162.** A context-free grammar is said to be ambiguous if
(A) it has useless productions (B) it generates an infinite language (C) some string has more than one parse tree (D) it is left recursive
- Q163.** The number of states in the minimal DFA for the regular expression $(0 + 1)^*11$ is
(A) 5 (B) 2 (C) 3 (D) 4
- Q164.** The intersection of a context-free language with a regular language is always
(A) not context-free (B) context-free (C) finite (D) regular

SECTION 7 — COMPILER DESIGN

- Q165.** Peephole optimization is performed on
(A) the parse tree (B) a small sliding window of target code (C) the source program (D) the symbol table
- Q166.** Which parser is the most powerful among the following?
(A) CLR(1) (B) LALR(1) (C) SLR(1) (D) LL(1)
- Q167.** Operator precedence parsing is a form of
(A) predictive parsing (B) recursive descent parsing (C) shift-reduce (bottom-up) parsing (D) top-down parsing
- Q168.** The DAG representation of a basic block is useful for
(A) tokenizing (B) common sub-expression elimination (C) register spilling only (D) parsing
- Q169.** The symbol table in a compiler is
(A) used by almost all phases of the compiler (B) created only during code generation (C) discarded after parsing (D) built only during lexical analysis
- Q170.** Assigning a real value to an integer variable without a cast is detected during
(A) semantic analysis (B) syntax analysis (C) code generation (D) lexical analysis
- Q171.** YACC is a tool that generates a
(A) assembler (B) lexical analyzer (C) parser (D) code optimizer
- Q172.** An S-attributed syntax-directed definition uses
(A) no attributes (B) both, evaluated left to right (C) only synthesized attributes (D) only inherited attributes

SECTION 8 — OPERATING SYSTEM

- Q173.** Which CPU scheduling algorithm is preemptive by design and gives every process a fixed time slice?
(A) Round Robin (B) SJF (C) FCFS (D) Priority (non-preemptive)
- Q174.** A race condition occurs when
(A) a process waits indefinitely for a resource (B) processes never share data (C) the final result depends on the relative order of accesses to shared data (D) CPU utilization reaches 100%
- Q175.** Thrashing in a virtual memory system is directly caused by
(A) large page size (B) contiguous allocation (C) an excessively high page fault rate (D) too few processes in memory
- Q176.** Which disk scheduling algorithm can cause starvation of requests located far from the current head position?
(A) SCAN (B) SSTF (C) C-SCAN (D) FCFS
- Q177.** In a system with a 32-bit logical address space and 4 KB pages, the number of pages in the logical address space is
(A) 2^{20} (B) 2^{10} (C) 2^{32} (D) 2^{12}
- Q178.** Which scheduling policy can cause indefinite postponement (starvation) of long jobs?
(A) First-in first-out with aging (B) Round Robin (C) FCFS (D) Shortest Job First

- Q179.** A counting semaphore initialized to 1 and used with wait/signal around a critical section behaves as
(A) a spin lock only (B) a condition variable (C) a monitor (D) a binary semaphore / mutex
- Q180.** A zombie process is one that
(A) has no parent process (B) is permanently blocked on a semaphore (C) is waiting for an I/O device (D) has terminated but whose exit status has not yet been collected by its parent
- Q181.** The problem of priority inversion in real-time systems is addressed by
(A) aging (B) priority inheritance (C) FCFS scheduling (D) round robin scheduling
- Q182.** In which threading model does the blocking of one thread NOT block the entire process?
(A) Pure user-level threads (B) Kernel-level threads (C) Neither model (D) Both behave identically
- Q183.** Belady's anomaly is exhibited by which page replacement policy?
(A) LFU (B) FIFO (C) LRU (D) Optimal
- Q184.** Four processes arrive at time 0 with CPU bursts 6, 8, 7 and 3 units. Under SJF scheduling the average waiting time is
(A) 10.25 units (B) 7 units (C) 6.5 units (D) 8 units
-

SECTION 9 — DATABASES

- Q185.** Which ACID property guarantees that the effects of a committed transaction survive subsequent system failures?
(A) Consistency (B) Atomicity (C) Durability (D) Isolation
- Q186.** Deadlock prevention in database concurrency control is achieved by schemes such as
(A) shadow paging (B) checkpointing (C) two-phase commit (D) wait-die and wound-wait
- Q187.** A trigger in a relational database is
(A) an index structure (B) a procedure executed automatically in response to a specified event (C) a constraint checked manually (D) a virtual table
- Q188.** Which of the following is a DDL command?
(A) DELETE (B) UPDATE (C) CREATE (D) SELECT
- Q189.** A primary index built on the ordering key field of an ordered data file is normally
(A) a secondary index (B) always multilevel (C) sparse (D) dense
- Q190.** R(A, B, C, D) has the FD set {AB → C, C → D, D → A}. The highest normal form satisfied by R is
(A) 2NF (B) BCNF (C) 3NF (D) 1NF
- Q191.** For R(A, B, C) with the FD set {A → B, B → C}, the candidate key is
(A) A (B) C (C) BC (D) B
- Q192.** A relation is in first normal form (1NF) if
(A) every attribute value is atomic (B) it has no transitive dependency (C) every determinant is a candidate key (D) it has no partial dependency
-

SECTION 10 — COMPUTER NETWORKS

- Q193.** Cyclic Redundancy Check (CRC) is used for
(A) encryption (B) routing (C) flow control (D) error detection
- Q194.** The number of usable host addresses in the subnet 192.168.10.0/26 is
(A) 62 (B) 126 (C) 30 (D) 64
- Q195.** IEEE 802.11 wireless LANs use which medium access method?
(A) Pure ALOHA (B) CSMA/CA (C) CSMA/CD (D) Token ring
- Q196.** The length of an IPv6 address is
(A) 64 bits (B) 128 bits (C) 256 bits (D) 32 bits
- Q197.** Which of the following normally uses UDP at the transport layer?
(A) HTTP (B) DNS query (C) SMTP (D) FTP control connection
- Q198.** Which transport layer protocol is connectionless?
(A) FTP (B) TCP (C) SCTP (D) UDP
- Q199.** The ping utility works using which protocol?
(A) IGMP (B) ARP (C) SNMP (D) ICMP
- Q200.** Fragmentation of IP datagrams is performed at which layer?
(A) Application layer (B) Transport layer (C) Network layer (D) Data link layer
-

ANSWER KEY — MOCK CBT 1

Paper-I (Q1 - Q100)

1-C 2-A 3-A 4-D 5-B 6-D 7-A 8-B 9-D 10-B
11-A 12-C 13-A 14-B 15-B 16-B 17-D 18-B 19-B 20-A
21-A 22-D 23-D 24-B 25-D 26-D 27-B 28-B 29-B 30-B
31-C 32-C 33-B 34-A 35-D 36-B 37-A 38-A 39-D 40-B
41-C 42-C 43-D 44-D 45-B 46-B 47-B 48-A 49-C 50-D
51-C 52-D 53-B 54-A 55-A 56-A 57-B 58-C 59-C 60-D
61-D 62-A 63-A 64-C 65-A 66-A 67-D 68-A 69-D 70-D
71-A 72-D 73-C 74-A 75-D 76-A 77-A 78-D 79-A 80-D
81-C 82-D 83-B 84-A 85-D 86-A 87-B 88-B 89-C 90-A
91-D 92-B 93-C 94-A 95-B 96-C 97-B 98-B 99-B 100-B

Paper-II (Q101 - Q200)

101-C 102-C 103-C 104-B 105-A 106-A 107-A 108-C 109-D 110-A
111-D 112-C 113-B 114-D 115-A 116-D 117-A 118-C 119-B 120-A
121-B 122-B 123-C 124-D 125-B 126-C 127-A 128-B 129-B 130-D
131-A 132-A 133-A 134-B 135-C 136-A 137-A 138-C 139-A 140-A
141-A 142-C 143-D 144-D 145-D 146-A 147-A 148-D 149-D 150-A
151-C 152-C 153-A 154-B 155-D 156-C 157-C 158-B 159-A 160-B
161-B 162-C 163-C 164-B 165-B 166-A 167-C 168-B 169-A 170-A
171-C 172-C 173-A 174-C 175-C 176-B 177-A 178-D 179-D 180-D
181-B 182-B 183-B 184-B 185-C 186-D 187-B 188-C 189-C 190-C
191-A 192-A 193-D 194-A 195-B 196-B 197-B 198-D 199-D 200-C

Score sheet

Paper-I: ____ / 100 (qualifying: 40 UR/EWS, 35 OBC-NCL, 30 SC/ST/PwD)

Paper-II: ____ / 100 (same qualifying marks)

Total: ____ / 200

Section-wise wrong answers — GA ____ Reasoning ____ Numerical ____ English ____ | Engg Maths ____ Digital Logic ____ COA ____ Prog & DS ____
Algorithms ____ TOC ____ Compilers ____ OS ____ DBMS ____ CN ____